

# Review Comments on “Coordinated Beamforming for Networked Integrated Communication and Multi-TMT Localization”

The paper investigates coordinated beamforming for networked ISAC systems where dedicated TMTs, rather than BSs, handle sensing. It derives a closed-form CRLB for ToA-based localization and proposes both optimal and SCA-based algorithms for sensing-centric and communication-centric designs. Simulations reveal clear communication–localization trade-offs and show that adding more TMTs yields greater gains than adding more BSs.

Based on my reading of the manuscript, I have the following comments on the paper:

1. The statement in the abstract “To further enhance networked ISAC, delegating the reception of sensing signals to dedicated target monitoring terminals (TMTs) instead of base stations (BSs) offers significant advantages in terms of sensing capability and deployment flexibility” remains misleading because it suggests that using TMTs rather than BSs for sensing is a discretionary design choice, whereas it effectively reflects a shift from monostatic to bistatic sensing and should not be presented as an optional enhancement.
2. The abstract does not clearly justify why the study focuses on separate sensing-centric and communication-centric designs while omitting a joint design strategy, leaving the underlying motivation for this separation insufficiently explained.
3. The paper motivates its architecture by drawing parallels to multi-BS cooperative communication, yet such cooperation has seen limited practical deployment due to synchronization and backhaul constraints. The authors should clarify why multi-BS–cooperative sensing/ISAC would be any more feasible in practice, as this gap weakens the justification for the proposed framework.
4. Although the manuscript notes that prior ISAC work has largely focused on target detection and parameter estimation rather than target localization, it does not articulate the unique challenges associated with localization. The authors should briefly explain why localization is fundamentally more difficult, particularly how it differs from parameter estimation in terms of beamforming design, geometric constraints, and the structure of the CRLB. Without this clarification, the motivation for treating localization as a distinct and understudied problem remains incomplete.
5. The manuscript assumes perfect clock synchronization between BSs and TMTs, which is critical for ToA-based localization. In practice, synchronization errors or timing offsets can significantly degrade ToA accuracy and, consequently, the CRLB and the resulting beamforming design. The authors should comment on the impact of synchronization mismatch and clarify whether their framework can accommodate such imperfections or how sensitive the proposed methods are to synchronization errors.
6. The notations paragraph states that  $\|\cdot\|$ ,  $|\cdot|$ ,  $\text{tr}(\cdot)$ , etc. represent the Euclidean norm, the modulus, the trace, and so on, of matrices or vectors. The term ‘modulus’ is ambiguous in this context since modulus is typically defined only for scalars. The authors should clarify what  $|\cdot|$  denotes for matrices or vectors, or revise the notation to avoid confusion.”
7. The assumption that TMTs can be deployed near the target in a distributed manner raises concerns regarding the practicality of the system model:

- (a) The response letter describes TMTs as movable IoT devices. It is unclear what specific devices these are, how they would be repositioned in practice, and who bears the cost of deploying or relocating them for sensing a single target.
  - (b) In multi-target scenarios, it is not clear how many TMTs would need to be moved closer to each target or how the optimal spatial configuration should be determined. The authors should clarify how such deployment decisions are made and discuss the associated overhead.
  - (c) The manuscript states that clock synchronization between BSs and TMTs can be established via wired links. If TMTs are indeed movable IoT devices, it is unclear how wired connections would be provided or maintained. The authors should reconcile the assumption of mobility with the requirement for wired synchronization.
8. Since no dedicated sensing signals are transmitted from the BSs, it is unclear how reliable sensing or localization is achieved using communication-only waveforms. Do the authors implicitly assume that the transmit beamwidth is sufficiently wide to illuminate the target? If so, this may lead to inefficient beamforming, and it is not clear how the system performs when the target lies at a large angular separation from the intended communication users. The authors should clarify the underlying assumption and discuss its implications for sensing performance.
  9. The assumptions of perfect CSI for all communication users and a clutter-free, NLoS-free environment for target sensing are overly optimistic and far from practical operating conditions. The authors should discuss how the proposed methods would be affected under more realistic channel uncertainties and sensing environments.
  10. While the authors note that gradient-based methods are unsuitable in their pure form, both problems (15) and (16) can still be addressed using a penalty-based reformulation followed by a gradient-based algorithm to obtain a stationary solution. Although the proposed SDP-based method provides globally optimal solutions, which is commendable, its practical utility is limited due to computational complexity. It would be valuable for the authors to comment on or investigate the optimality gap between their solutions and those obtained via a penalty-aided gradient approach. Note that the discontinuity in (16a) can be handled via any of the suitable smoothing approximation techniques.

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11. The multi-target scenario raises several concerns that are not addressed in the manuscript:
  - (a) For widely separated targets, the paper assumes separate groups of TMTs for each target, but it does not explain how optimal TMT-to-target association is determined.
  - (b) For closely spaced targets, inter-target reflections and mutual interference cannot be ignored. It is unclear whether the current framework accounts for such coupling effects. The authors should clarify how these issues are handled or discuss the limitations of the model.
12. The authors should include a single-BS benchmark to concretely demonstrate the benefits of multi-BS cooperation. For a fair comparison, the total transmit power budget should be kept identical in both cases. This would help quantify the actual performance gain attributable to multi-BS cooperation.
13. The power constraint in the manuscript is imposed per BS, so when the number of BSs increases, the total network transmit power also increases. This makes comparisons across different values of  $M$  potentially unfair, as performance gains may partly stem from a larger total power budget rather than from cooperation itself. The authors should either normalize the total transmit power across different BS configurations or provide additional discussion to justify the current comparison.